

Problem Set 6 Solutions

QTM 220

Due 1 December at 2:00pm ET on Canvas. Questions about submission requirements should be directed to your TA.

California CPS Revisited

- (a) Using the California sample of 25-35 year olds, regress income on education and age (in an additive model). Report your results in a table, including the standard errors that R automatically calculates.

```
cps<- read.csv("California_CPS_degrees.csv")

mod1<- lm(income~ education + age,data=cps)

summary(mod1)
```

Call:

```
lm(formula = income ~ education + age, data = cps)
```

Residuals:

Min	1Q	Median	3Q	Max
-226476	-62493	-20343	36193	752685

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-137806.3	24024.5	-5.736	1.1e-08 ***
education	16479.0	889.8	18.520	< 2e-16 ***
age	1051.6	694.4	1.514	0.13

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Residual standard error: 101800 on 2243 degrees of freedom

Multiple R-squared: 0.1345, Adjusted R-squared: 0.1337

F-statistic: 174.2 on 2 and 2243 DF, p-value: < 2.2e-16

- (b) Create a matrix of your X information for your model (education and age) and include a column of 1s to estimate an intercept. Using the following matrix functions in base R, calculate $(X'X)^{-1}X'Y$ and confirm your resulting $\hat{\beta}$ s are the same as your *lm* results:

Let X be your matrix in R. Use `%*%` to multiply matrices, i.e. `X%*%X` will yield XX

Use `t(X)` to take the transpose of matrix X

Use `solve(X)` to take the inverse of matrix X

```
df.mat<- data.frame(constant=rep(1,length(cps$income)), edu=cps$education, age=cps$age)

X<- data.matrix(df.mat)

solve(t(X)%*%X)%*%t(X)%*%(cps$income)
```

```
      [,1]
constant -137806.340
edu       16478.997
age       1051.582
```

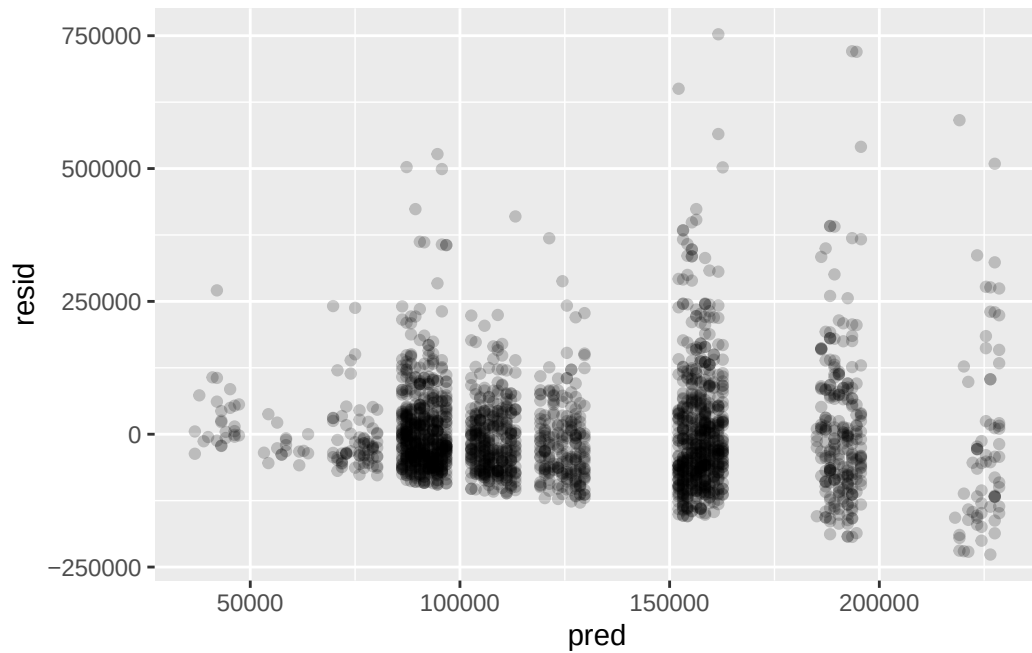
- (c) Extract the predicted values and residuals from your model using `predict(model name)` and `residuals(model name)`, then create a scatter plot of your predicted values and residuals. What do you see? Would you guess the errors are homoskedastic or heteroskedastic? Why?

```
pred<- predict(mod1)
resid<- residuals(mod1)

df<- data.frame(pred=pred,resid=resid)

rplot<- ggplot(df, aes(x=pred,y=resid)) + geom_point(alpha=.2)

rplot
```



- (d) install the sandwich package and get heteroskedasticity consistent standard errors for your model
- (e) do matrix algebra to get the same thing

Neto and Cox

Propensity score matching

make a balance table

Menopausal Hormone Therapy

compare RCT and observational studies

read some stuff

reflect